

STATS 503, Homework 9

1. **[10 POINTS]** ISLP chapter 12 exercise #2
2. **[6 POINTS]** Suppose that you have a dataset with two predictors and perform PCA by carrying out an eigen-decomposition on the covariance matrix of the data (the two predictors are on the same scale). You find that its two eigenvalues are $\lambda_1 = 4$ and $\lambda_2 = 1$, and their corresponding eigenvectors are

$$u_1 = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} \quad u_2 = \begin{bmatrix} -0.8 \\ 0.6 \end{bmatrix}$$

- (a) Reconstruct the covariance matrix of the data. *Hint: recall that the eigendecomposition of a real **symmetric** $p \times p$ matrix T can be written as $T_{i,j} = \sum_{k=1}^p \lambda_k u_{k,i} u_{k,j}$ where $\lambda_1, \dots, \lambda_p$ are the eigenvalues and $u_1, u_2 \dots u_p$ are the corresponding orthonormal eigenvectors.*
 - (b) What percentage of variance is explained by the first principal component?
 - (c) For a new data point $X = (1, 2)$, find its scores on the first and second principal components.
3. Consider the `arrests` data, which can be downloaded from canvas.
- (a) **[2 POINTS]** Using hierarchical clustering with complete linkage and Euclidean distance, cluster the states. Plot the dendrogram.
 - (b) **[2 POINTS]** Cut the dendrogram at a height that results in five distinct clusters. Report the states belonging to each of the five clusters. Make a silhouette coefficient plot.
 - (c) **[2 POINTS]** Perform K-means clustering on the data with $K = 5$ and report which states belong to which clusters. Report how you initialized the algorithm. Make a silhouette coefficient plot and comment on any interesting features.
 - (d) **[6 POINTS]** Scale the variables to have mean 0 and standard deviation 1. Repeat questions (a)–(c) using the scaled data.
 - (e) **[2 POINTS]** Run PCA on the dataset with 1, 2, and 3 PCs. Report the variance explained for each of the three PCA fits.

Total points for this assignment: 30